



TEXAS A&M UNIVERSITY

School of Engineering Medicine

MitraClip Anatomy-Informed Ultrasound Repair Assistance MC-AURA

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IIE 2026

Clinical Problem

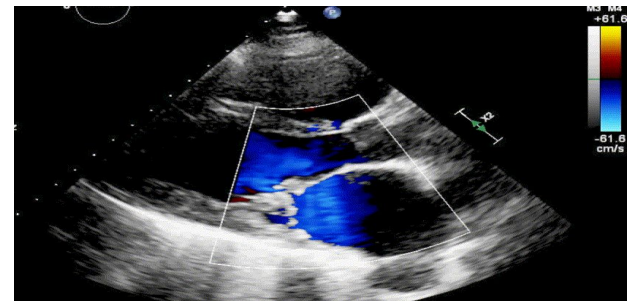
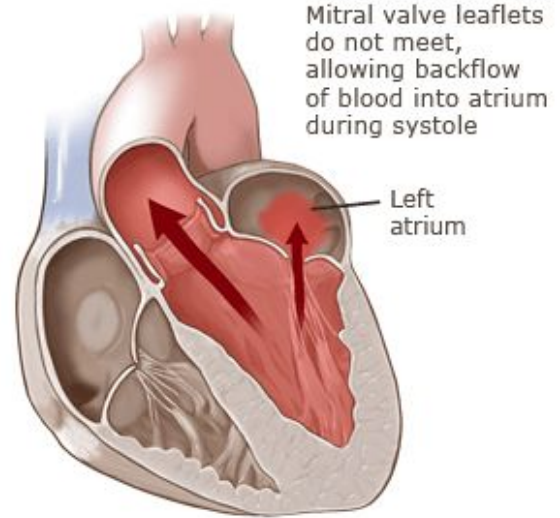
Mitral Regurgitation: “unwanted backflow of blood from the left ventricle to the left atrium of the heart via a “leaky” mitral valve”

MR affects **1 in 10** adults over 75, one of the most common valve diseases in the U.S.

Causes

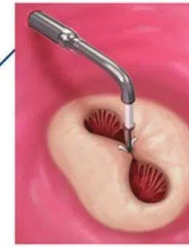
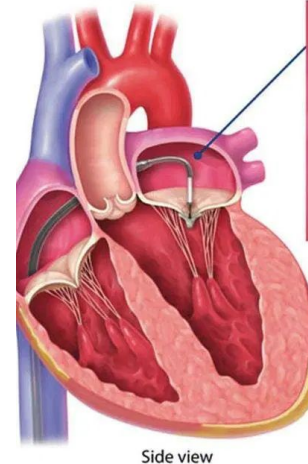
Primary (structural) Secondary (functional)

Mitral Regurgitation



Current Solutions for MR

- Open heart/minimally invasive surgery to replace the mitral valve
- Transcatheter Mitral Valve Replacement (TMVR)
- Transcatheter Edge to Edge Replacement (TEER)



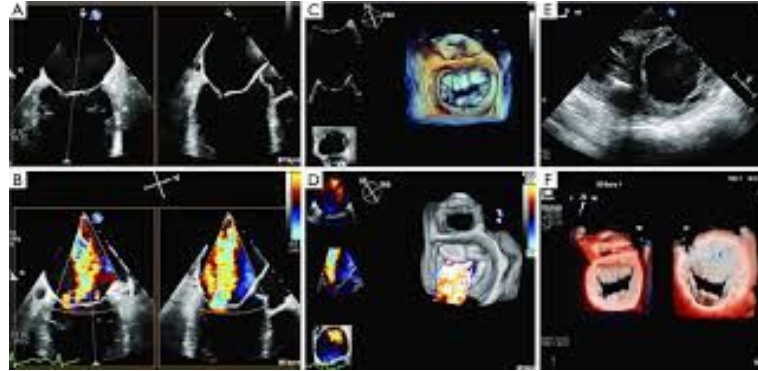
Atrial view

**MitraClip: Transcatheter Edge-to-Edge Repair (TEER)
for Mitral Regurgitation**

YASHODA
HOSPITALS



Need Statement

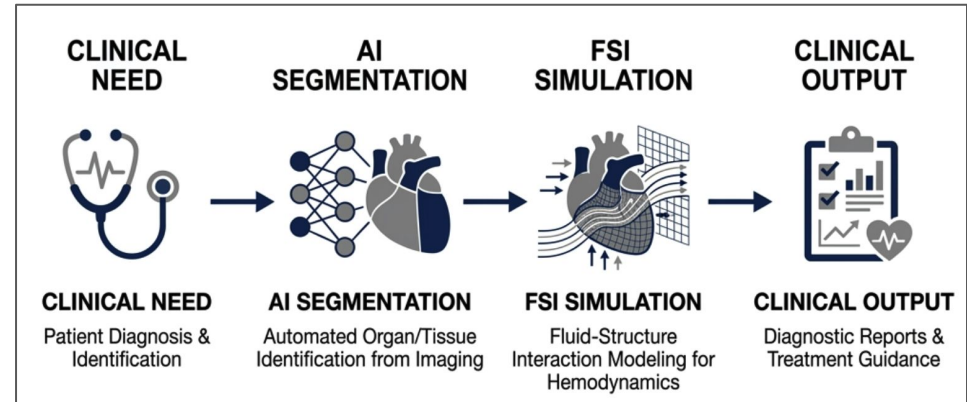


There remains a need for a way to help clinicians determine the **optimal mitral clip positioning** to **decrease** the proportion of patients that show **more than residual MR post-treatment**.

Learning Outcomes

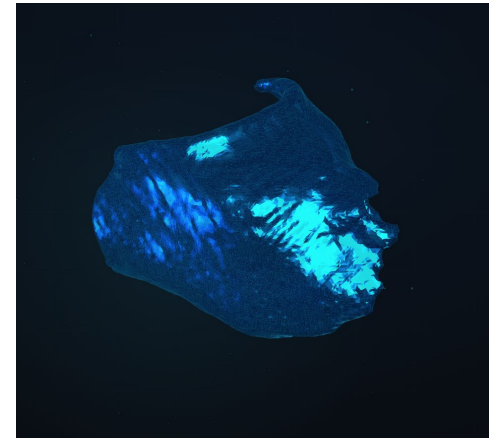
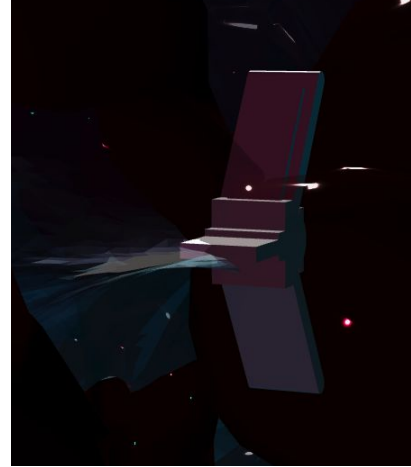
Pursuing IIE Learning Outcomes 3 and 4.

- **LO 3 Goal (Develop Concepts):**
Translate clinical pain points into formal software design inputs to engineer a computational pipeline.
- **LO 4 Goal (Evaluate Prototypes):**
Utilize segmented 3D TEE data to test patient-specific fluid-structure interaction (FSI) frameworks.



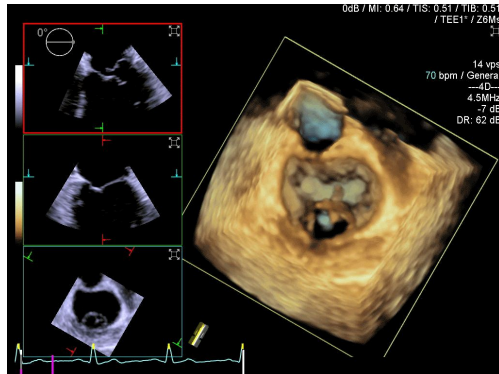
LO Methods

- 1) **MVP Research:** Evaluated several AI models and FSI simulations to select the best tools suited for our problem
- 1) **Valve Segmentation / Meshing:** Extracted anatomical reconstructions directly from 3D TEE volumes and created meshes for flow simulation
- 1) **Flow Simulation / Testing:** Ran virtual clip configurations in silico to model flow and evaluate mechanical stress and regurgitant volume reduction

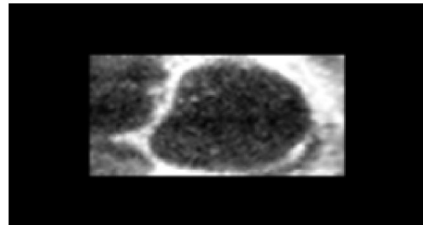


Segmentation and Meshing

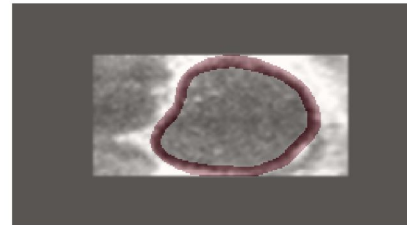
- Extracted non-zero volumetric data from clinical TEE scans as per the MVSeg2023 Database, bypassing manual geometric construction.
- Generated high-fidelity, repaired, and volumized PyVista meshes for distinct patient geometries.
- **Impact:** Automates the geometric modeling phase, validating technical feasibility for real-time clinical deployment.



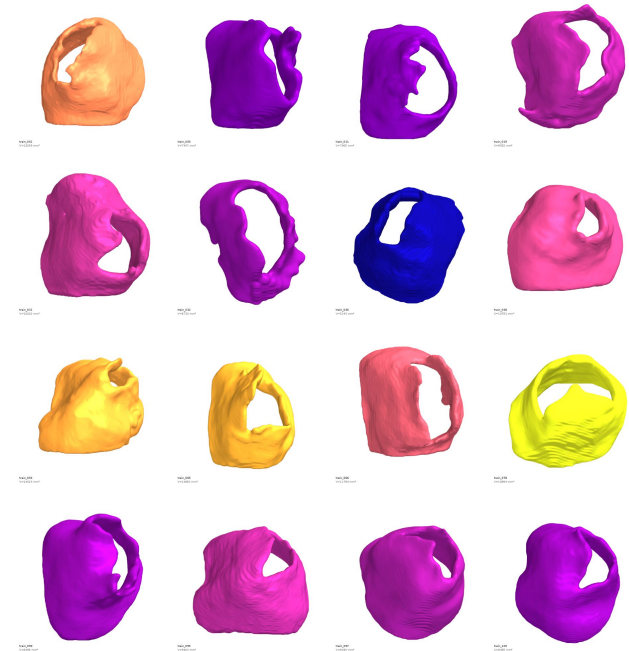
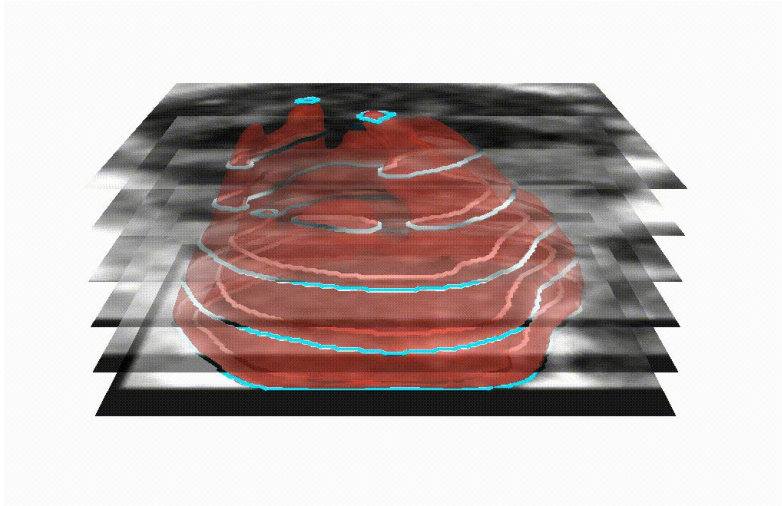
Ultrasound (slice 17)



Labeled ultrasound

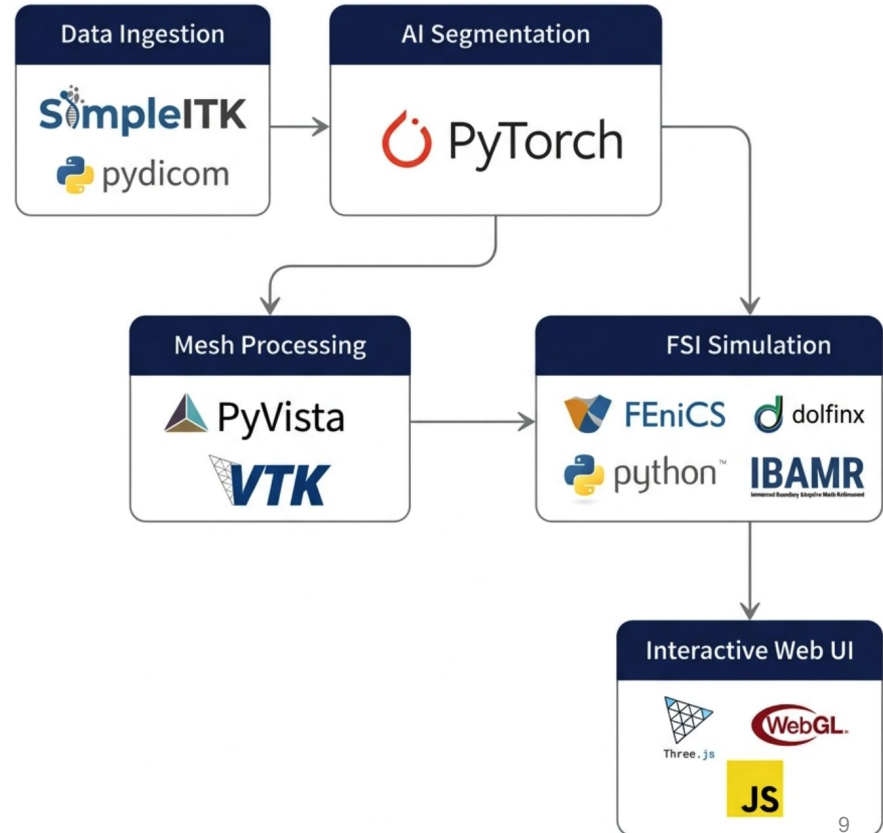


Segmentation and Meshing Outputs



FSI Model

- Successfully completed the Preliminary Navier Stokes Flow Simulation and finalized NTW/NT model architectures.
- Identified critical stability issues in the FSI model at peak systolic pressures.
- **Technical Pivot(s):** Tested out multiple FSI models (SimVascular, IBAMR, FEniCS/DolphinX). Actively developing V2 model utilizing FEniCS and DolphinX computational environments to ensure convergence

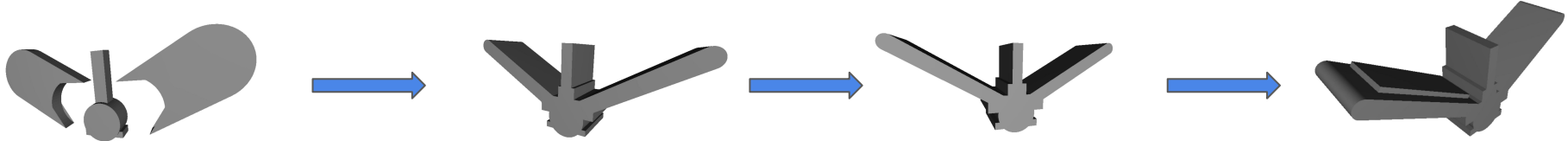




Flow Simulation

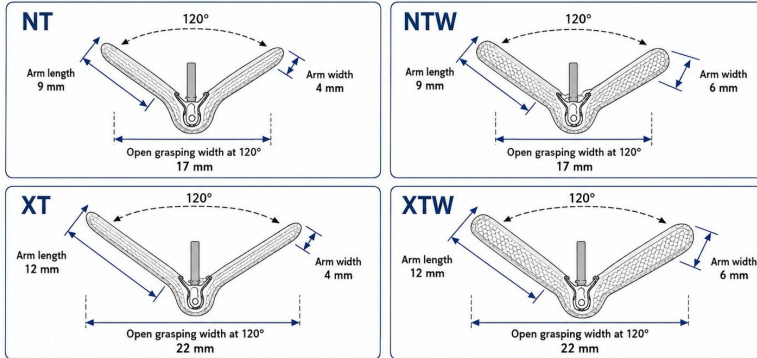


Designing the Model MitralClip



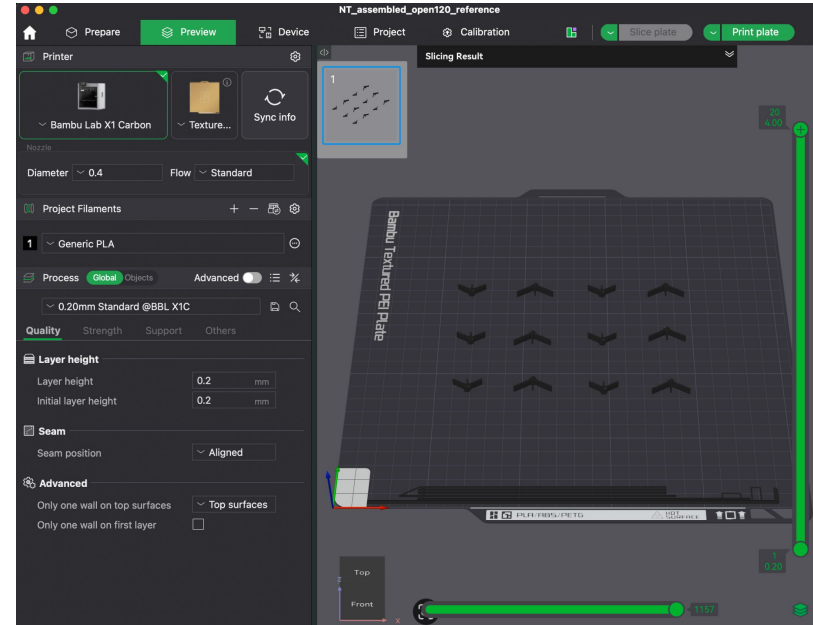
MitraClip G4 Size Comparison

Conceptual dimensional mockup for planning discussion



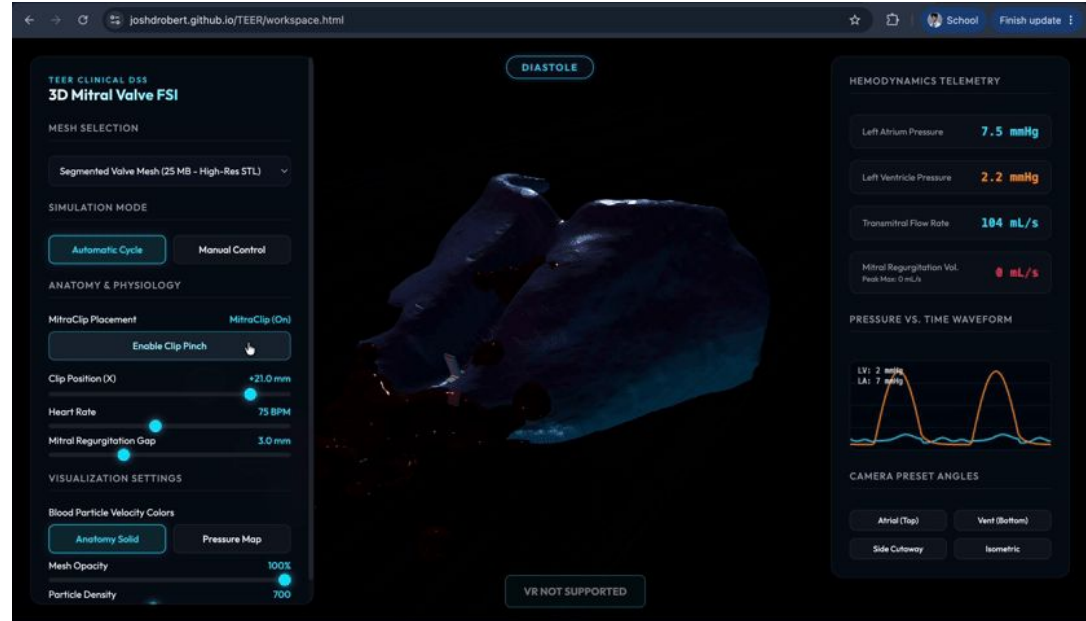
Model	Arm length	Arm width	Open width at 120°
NT	9 mm	4 mm	17 mm
NTW	9 mm	6 mm	17 mm
XT	12 mm	4 mm	22 mm
XTW	12 mm	6 mm	22 mm

Schematic based on published G4 size dimensions; for educational discussion only.



Final Results

- Tested 1, 2, and 3 clip deployments across distinct spatial zones in silico.
- Deployed a functional, web-based interactive demonstration of the MC AURA tool. Adapted VR functionality.
- **Impact:** Provides the verified training dataset necessary for the downstream machine learning optimization algorithm to predict outcomes instantly.



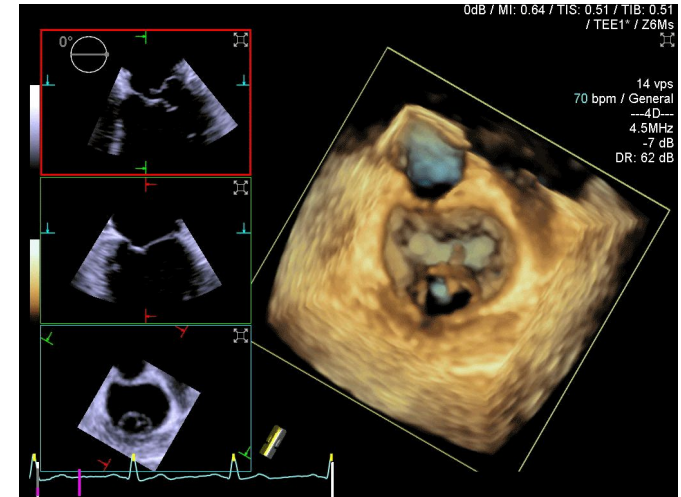
Flow Simulation [Live Demo]



<https://joshdrobert.github.io/TEER/>

Next Steps

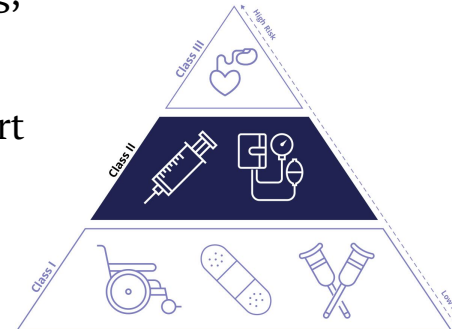
- Adapt current FSI model to pre-existing validated model / parameters. Continue refining to match physiology
- Build FSI models for functional pathologies that cause MR
- Segment both valve structure and blood flow from TEE videos to incorporate each patient's anatomy **and** physiology
- Interview clinicians who regularly perform TEER procedure (early and late stage of practice). Co-design features and functionality of the user-facing platform.



Future Work

- VR Pathway: Initiated the development of a virtual reality adaptation specifically designed for seamless integration with KARL STORZ surgical visualization hardware
- Administrative: Finalize formal Institutional Review Board data-sharing proposal with Houston Methodist Hospital to acquire matched TEE and outcome data.
- Validation: Conduct physical phantom benchtop testing, followed by in vivo animal validation.
- Regulatory: Pursue FDA Class II clinical decision support software classification.

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Appendix Slides



Acknowledgements

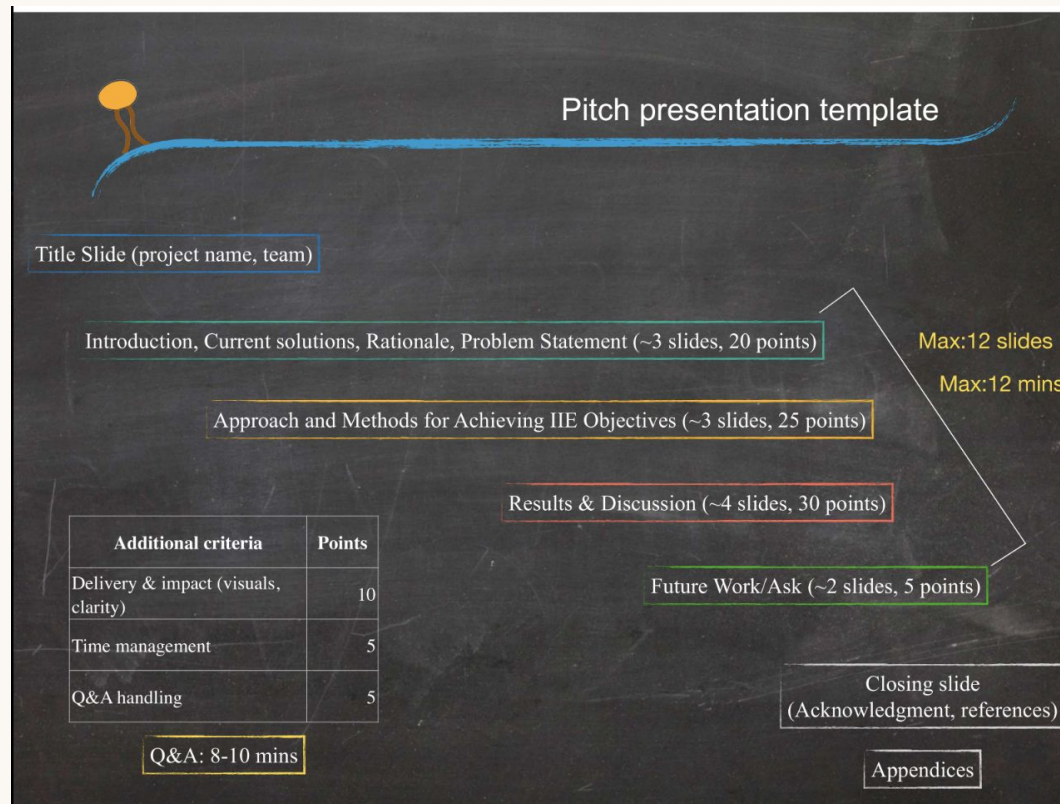
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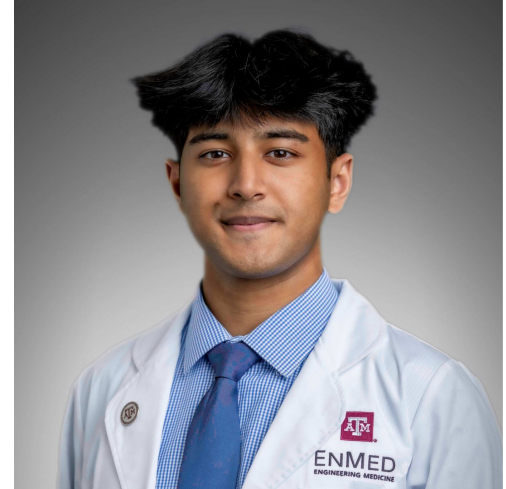
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